

SYNOPSIS



Two powerboats collided during a race at the Dover Regatta causing fatal injuries to one crew member.

Both crews had limited racing experience, but were taking part in the Royal Yachting Association's (RYA) national championship, on a course that was too short and was congested by the number of boats in the race. The collision occurred when boat A89 unintentionally lost control, turned sharply and reduced speed significantly. Boat A2, following behind, had little opportunity to take avoiding action and struck the side of A89 close to where its co-driver was seated. The

co-driver of A89 bore much of the force of the impact, his helmet was torn off and he suffered severe injuries to his head and upper body. Despite prompt and extensive medical attention he could not be saved.

The event took place on a foreshortened and compromised course, under confusing rules and without the risks being properly assessed. However it was approved by the RYA, and a large proportion of untrained novice and inexperienced competitors were permitted to race. Ultimately, it was the ability of the crews of the two boats that collided which caused the collision. However, they were racing under the auspices of an organisation that the investigation concluded had been insufficiently focused on safety, and had not made adequate attempts to control the race, or educate the crews about the risks they faced.

Following its investigation into a junior, K-Class powerboat racing accident at Portland Harbour on 19 June 2005, the MAIB recommended the RYA should:

Consider the safety issues arising from this accident, and develop a pro-active safety management system which is subject to an independent audit by a professional body, to ensure effective oversight of powerboat racing. Particular attention should be given to developing procedures for the oversight of the K-Class racing classes.

The RYA accepted the recommendation, however, the investigation found that the safety lessons identified in the K-Class accident had not been applied effectively across the sport, as intended by the RYA Council. This had allowed a number of systemic weaknesses to persist, which set the pre-conditions for the accident at Dover.

Following the MAIB's investigation, the RYA Council has affirmed its ownership of safety including the need to provide clear guidance and oversight of safety issues at all levels of the RYA. Within the RYA this will include: providing subordinate committees and structures with defined responsibilities, authority and accountability; promoting a culture of continuous improvement in safety; and enhancing auditing and monitoring of safety.

In light of the actions agreed by the RYA Council, this report makes no safety recommendations.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF VESSELS AND ACCIDENT

Sleepwalker

Race no	:	A2
Built	:	2006
Design	:	Concord
Construction	:	Glass Reinforced Plastic
Measured length	:	5.50m
Maximum beam	:	1.59m
Minimum weight	:	525kg
Engine power and type	:	Yamaha V4,130hp, 2 stroke outboard engine
Maximum speed	:	~ 60mph

Harwich 2011

Race no	:	A89
Built	:	2005
Design	:	Phantom
Construction	:	Glass Reinforced Plastic
Measured length	:	6.03m
Maximum beam	:	1.78m
Minimum weight	:	525kg
Engine power and type	:	Yamaha V4,130hp, 2 stroke outboard engine
Maximum speed	:	~ 60mph

1.2 BACKGROUND

Offshore circuit racing is a popular style of racing for powerboats. The course is typically close to land or within harbour limits, and each turning point in the circuit is marked by a buoy. The racers complete a number of laps, and in this case the total length was determined by a 25 minute time limit. Various different classes of boats take part in this style of racing, but both the boats involved in this accident were racing in the Offshore Circuit Racing (OCR) 'A' class. The race was part of the Royal Yachting Association's (RYA) national championships, and the RYA has conducted its own investigation into the accident.

1.3 NARRATIVE

Powerboat racing had been arranged by the UK Formula Future Offshore Racing Club (UKFFORC) as part of the annual Dover Regatta, which took place during the weekend of 8 and 9 August 2009. Two heats were planned for each day of the regatta. The collision took place towards the end of the first powerboat race of the event.

It was reported by racing clubs and the RYA that the numbers of competitors in powerboat racing had been in decline, and different classes were encouraged to race together to share costs and keep events viable. OCR class boats joined in with UKFFORC at Dover to share the organisation and infrastructure of racing officials, safety crews and insurance overheads.

1.3.1 Offshore circuit racing classes

Two classes of UKFFORC and two classes of OCR boat competed in the same race at Dover, for points in their own respective championships. The majority were OCR 'A' class boats, which are between 5.5 and 6.1m (18 and 20 feet) in length and powered by a 1.8 litre, 2 stroke outboard engine. The class has existed for many years, and is regarded as being a relatively accessible way of taking part in the sport. There was a single 'B' class boat racing which was almost the same in appearance, but fitted with a smaller, 1.3 litre engine. 'A' and 'B' classes have traditionally raced together on the same courses.

The UKFFORC runs a number of classes for junior racers, but offers a progression path to adult racing with E900 and the E1500 classes. Both these were outwardly similar to OCR 'A' and 'B' boats, with the major difference being the use of low exhaust emission type 2 stroke engines, of either 90 or 150hp output respectively. The E1500 was in its first year, as an experimental class, and this was the first national event where OCR 'A' and E1500 boats had raced together on the same course.

At the beginning¹ of the race, there were 14 boats from four different classes competing:

- 3 x E1500 class boats
- 2 x E900 class boats
- 8 x OCR 'A' class boats
- 1 x OCR 'B' class boat

1.3.2 Racing circuit

The course was designed by UKFFORC and used eight buoys to make an oval shape, modified on two sides with chicanes (**Figure 1**). The first turn consisted of two buoys (no.s 1 and 2), to create a longer arc and encourage separation between the boats at the end of the starting run. This feature was introduced to racing after a collision between two boats in a junior class which occurred at the first corner of a race in 2005². The first chicane on the back straight was relatively gentle, but prevented the boats from straying too close to the beach. Buoy no. 5 was positioned towards the Prince of Wales Pier and was set out from the natural oval shape. It was followed by a chicane which made the turn around buoy no. 5 acute and caused the boats to slow as they approached the pier. The chicane then led the drivers through a more challenging right hand turn past buoys 6 and 7. The accident occurred near the last turn at buoy no. 8, a 90° left hand turn back onto the main starting straight.

1.3.3 Collision

Three separate video recordings of the race, taken from different vantage points on the Prince of Wales Pier, were studied to examine the race. MAIB contracted a world championship winning powerboat racer to advise on racing techniques and assist with interpretation of events and analysis of the collision.

The race was started at about 1132, and the allotted 25 minutes elapsed as the leaders completed their 16th lap, and the last lap flag was flown. One of the videos of the race showed the collision and events immediately beforehand, from which it was possible to determine the courses followed by the boats (**Figure 2**). Boat B27 approached buoy no. 8 and took a tight line around the turn. Boat A28 was behind, but catching up and preparing to overtake (**Figure 3**). It followed a similar, but wider line to B27 around the turn keeping clear, but leaving it in position to overtake on the next straight. Timekeepers recorded that boat A89 was 2 seconds behind A28 and, from the video, there appears to be about one boat length between them as they approached buoy no. 8 (**Figure 4**). Boat A89 took an even wider line than A28, but during the turn, the driver attempted to cut inside of A28, in order to gain an advantage on the next straight. During this manoeuvre, A89 turned more sharply than was necessary, the bow trimmed down and the boat slowed significantly (**Figure 5**).

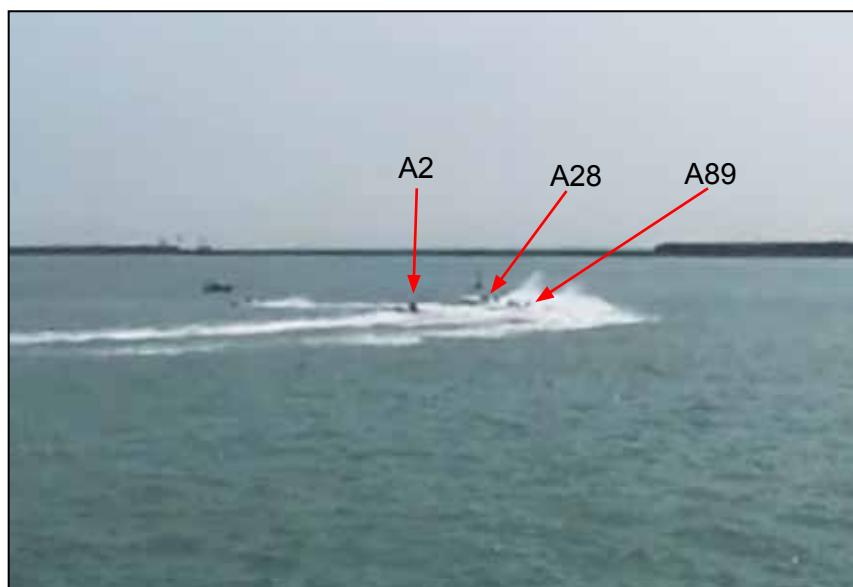
¹ Two boats had retired from the race due to equipment breakdowns by the time of the accident

² Collision between two Sorcerer powerboats at a junior K-Class event at Portland Harbour in June 2005 – [MAIB report 13/2006](#) published March 2006

Boat A2 was several boat lengths behind A89, but closed the distance quickly as it took a tight line into the turn at buoy no. 8 (**Figure 6**). Boat A2 began to turn and its crew saw A89 directly ahead, lying across their path. A2 struck A89 on the port side, in the part of the boat where the crew were seated. A2 was launched upwards and out of the water (**Figure 7**). As it crossed over A89, it struck A89's starboard side coaming immediately behind the driver's seat, and rolled to starboard. The driver of A2 was thrown clear, but the co-driver remained in the boat as it landed upside down in the water. After some initial difficulty, the co-driver swam clear.

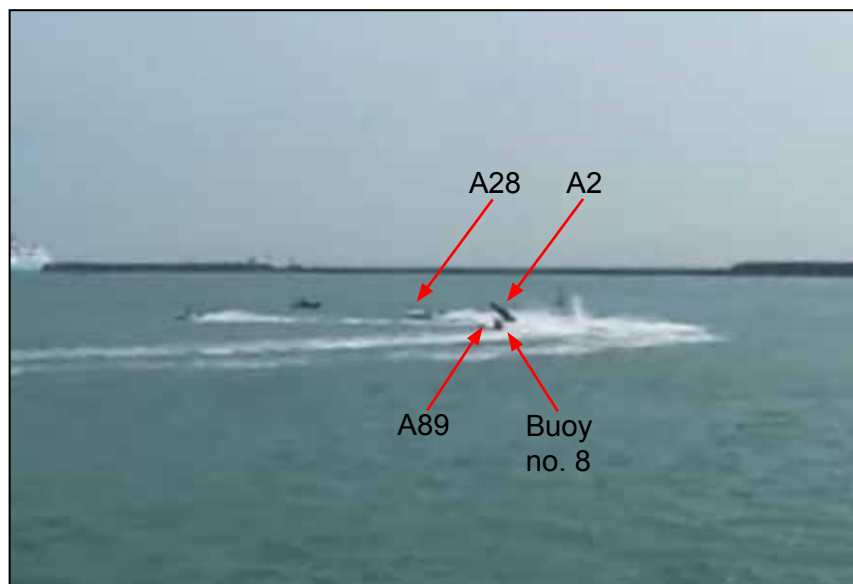
Examination of a GPS navigation unit taken from boat A89 showed that the collision happened at approximately 1157. On previous laps A89 had achieved speeds of about 45mph around buoy no. 8.

Figure 6



Boat A2 approaching the turn at buoy no. 8

Figure 7



Collision between boats A2 and A89

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 CONTRIBUTORY FACTORS

The collision and subsequent fatal injuries to the co-driver of A89 were caused by a combination of factors that had been allowed to develop and become common practice in powerboat racing. These included the following:

- The crews involved had no formal race training and were still developing their skills to recognise and avoid dangerous situations.
- The combination of course design and the number of boats racing meant that each boat had less space to manoeuvre.
- The design of the boats offered very little protection to the crew in the event of a collision.

None of these had been identified during preparations for the event, despite all of them being illustrated as factors in previous accidents. Other serious shortcomings in the planning, approval and running of the event included:

- The race was approved by the RYA despite the fact that it did not meet many of the requirements laid down in PB1.
- The process for assessing the risks for the race was weak, and the consequent risk assessment was not challenged by the Race Approvals Committee. There were no mitigation measures considered to reduce the increased risk created by the failure to meet the PB1 requirements.
- The risk assessment had not been read by any of the senior officials present at the event.
- The combined rules, created and approved to allow the two classes to race together were confusing and did not promote safer driving.
- Several unnecessarily hazardous events occurred during the race that were tolerated by both competitors and officials.

The RYA Council's attempts at introducing a safety management system following the accident at Portland Harbour in 2005 were ineffective, and it was unable to effectively control the risks associated with OCR. Specifically, its safety management system did not detect the weaknesses in race planning listed above, and did not enforce its own existing rules for powerboat racing.

2.3 COLLISION

2.3.1 Risk of collision

Boat A89 took a wider line into the turn at buoy no. 8, giving the driver the opportunity of leaving the corner more tightly and potentially gaining a racing advantage. In doing so, the boat was affected by the wakes of boats B27 and A28. The bow of A89 dropped, the boat slowed down and turned more tightly than was needed to follow the course, consistent with it having suffered a moderate hook. Boat A2 approached buoy no. 8 expecting boat A89 to have cleared the turn. The crew of A2 were not able to detect that A89 had hooked, and that there was a risk of collision in sufficient time to take effective avoiding action.

There is no evidence to indicate that any equipment in either boat failed prior to the collision.

2.3.2 Training and ability

The competitors in A89 and A2 had previous experience in operating fast boats, but they were relatively new to powerboat racing and had not done any formal race training. It was evident during the investigation that operating powerboats in a race on a short course needs different knowledge and skills than those required to drive a powerboat for work or leisure. Neither of these crews could be expected to perform to the same level as an experienced racing team, but they were permitted to race on a demanding course in a national championship. Any of the less experienced crews that were taking part could have found themselves being involved in a similar accident.

2.3.3 Crew protection

OCR rules do not require boats to have any additional measures to protect the crew from a collision. By plotting a line through the damaged area on A89 it can be seen that Mr Edmonds would have borne much of the force from the impact. However, both seats were damaged and the driver was extremely fortunate to have escaped without injury (**Figure 12**). All the boats in the OCR and the UKFFORC classes had similar levels of protection to boat A89, and their crews would all have been equally vulnerable to severe injury in the same circumstances. While some other powerboat classes have introduced safety features in their boats to protect crew, the investigation found no evidence of any attempt to improve crew protection in OCR class boats prior to this accident.

2.4 NATIONAL GOVERNING BODY

As with all motorsports, some risk is part of the excitement of racing, but races should be organised and competitors equipped with knowledge and skills to ensure that accidents are avoided and major injuries are prevented. Indemnity forms obliged competitors to accept racing risks based on their own assessment. However, in the absence of any mechanism to equip them with

2.5 LEARNING FROM PREVIOUS ACCIDENTS

The similarity between this accident and the one in Portland Harbour in 2005 is significant. Both accidents occurred during a moderate left hand turn during which the leading boat hooked and was subsequently struck by a following boat. Both accidents resulted in serious injuries to the leading boat's co-driver as he was hit by the following boat. Incident reports in 2005 and 2008 reinforce the likelihood of this type of accident. However, the PBRC, ORC and SAT committees, and the safety audit report, identified that incident reporting was poor and it is likely that many more cases went unrecorded. Given this and the subsequent lack of attention paid to incident reports by the senior committees, it is extremely unlikely that the RYA had a proper appreciation of the risks prevalent in the sport.

The changes made to K-Class racing after the accident at Portland demonstrate that the issues identified from the collision at Dover had already been considered several years before. Controlling separation between boats, training, and crew protection are common to both cases, and show that classes should not be treated in isolation. Lessons from accidents should, logically, be shared and implemented between classes in order to prevent future tragedies.

Tendencies to dismiss accidents or hazardous near misses as '*racing incidents*', implied that these events were accepted as routine hazards of normal racing and did not need to be considered further. One example of this, recorded in the minutes of an ORC meeting, clearly demonstrated a lack of understanding of the potential safety issues, no recognition of the safety role of the committee, and it also undermined the reporting and safety management system as a whole.

2.5.1 Previous recommendations to the RYA

Following the powerboat accident in Portland in 2005, the RYA Council unanimously accepted a recommendation from the MAIB to '*develop a proactive safety management system to ensure effective oversight of powerboat racing*'. While elements of a safety management system certainly existed, there was little evidence of the high level policy and focus needed to make the system work effectively. The PBRC had '*safety and rescue*' as a standing agenda item at its meetings, yet had not recorded any progress in this area and members of the committee had even questioned the requirement for this topic.

MAIB also recommended that the safety management system should be audited by an independent professional body to ensure effective oversight of powerboat racing. However, it took nearly 2 years from the RYA accepting this recommendation for the first internal safety audit report to be approved, and no external audits were commissioned. The first (internal) audit team was subsequently disbanded, and both safety audits that followed were completed by a past chairman of the PBRC. Both of his reports focused narrowly on established processes, and neither challenged the basic presumptions of safety in powerboat racing or the management systems.

As neither of these parts of the MAIB's recommendation was effectively implemented, a number of systemic weaknesses were allowed to persist and these set the pre-conditions for the fatal accident at Dover.

2.6 SAFETY MANAGEMENT IN POWERBOAT RACING

2.6.1 Course design and race approval

While it is not possible to show that the Dover course contributed directly to the collision, it is considered that the combination of course design and the number of boats racing put greater demands on the inexperienced racers taking part, and reduced safety margins.

Of the nine crews taking part in the OCR classes, six drivers and seven co-drivers had either provisional or event licences. RYA rules restricted competitors to two event licences each year, so these competitors must be considered to have been less experienced, or even novices. If the rule requiring them to mark their engines with a pink band had been enforced, officials would have had a highly visual reminder of the proportion of inexperienced racers that were participating. In its investigation, the RYA has since recommended that event licences should not be accepted for entry to national races.

Traditionally, novice OCR competitors learned their racing skills in 'basic' club races, which used simpler courses and had a minimum lap length of 2nm to allow each boat more space. A lap of the Dover course was a little over half this distance and, with the number of competing boats, it was inevitable that the racing would be close. While experienced racers might relish this challenge, and it provided good entertainment for spectators, it was not appropriate for the majority of crews who had limited experience.

At about 0.34nm, the start run was much shorter than the 1nm required by the rules and thus restricted the opportunity for the race fleet to spread out before the first corner. The start run also included a substantial navigation mark and, despite the course being modified by officials prior to the start, many boats passed close to it on both sides, therefore introducing an unnecessary risk of collision. The obstruction was not shown in the course approval application, nor was it discussed in race instructions.

Managing the different rules for a growing number of diverse classes was shown to be challenging, and maintaining and updating PB1 was time consuming. One of the participating clubs withdrew from assisting with organising the event after a disagreement about how the classes' rules should be applied, and the PRM and ORC chairman intervened a few days before the race to attempt to resolve the problem. However, implementation of the rules was still unclear. The race approval note quoted the rules incorrectly, and the PRM's decision that the UKFFORC rules should be used throughout became diluted to a confusing mix of rules from both classes.

SECTION 3 - CONCLUSIONS

The collision and fatal injuries to the co-driver of A89 were caused by a combination of factors that had been allowed to develop and become common practice in powerboat racing. The RYA's attempts to apply the lessons learned from the accident at Portland Harbour in 2005 were ineffective, and allowed systemic weaknesses in the management of powerboat racing to persist. These set the pre-conditions for this accident.

1. Boat A89 suffered a moderate hook while exiting a turn, and lay across the path of boat A2. The crew of boat A2 did not have enough time to recognise the risk of collision and take effective action to avoid it. [2.3.1]
2. Both of the crews involved had limited racing experience but were allowed to race on a demanding course in a national championship. Any of the less experienced crews that were taking part could have found themselves being involved in a similar accident. [2.3.2]
3. Crews operating the classes of boat used in this race are vulnerable to severe injury in a collision. [2.3.3]
4. The RYA did not have adequate policies to develop, apply and evaluate safety measures in powerboat racing. [2.4]
5. The collective effect of initiatives to encourage growth and participation in the sport eroded safety margins and created an environment where a serious accident was more likely. [2.4]
6. Both competitors and harbour authorities hosting events rely on the RYA's system of governance and race approval to give assurance that there is a safe framework in which racing can take place. This framework was significantly compromised at Dover. [2.4]
7. Lessons from previous accidents over several years have not been identified or shared effectively, preventing the risks in powerboat racing from being understood or addressed. [2.5]
8. Recommendations from the MAIB to the RYA following investigation of a similar accident have not been implemented properly and not as the RYA described at the time. This has allowed a number of systemic weaknesses in its safety management to persist. [2.5.1]
9. While it is not possible to show that the Dover course contributed directly to the collision, it is considered that the combination of course design and the number of boats racing put greater demands on the inexperienced racers taking part, and reduced safety margins. [2.6.1]

10. The RYA's race organisation and approval system had no way of assessing if the race was suitable for the inexperienced racers taking part; did not enforce the rules in PB1; did not properly reconcile rules for multiple classes; and took no action to mitigate the safety weaknesses in the proposed plan before approving it. [2.6.1]
11. Formal assessment of the risks during the organisation and preparation for the event was inadequate and illustrates weaknesses in safety management at every level of the association. [2.6.2]
12. The RYA's policy of only conducting detailed risk assessments for major events did not recognise that the risk to individuals remains similar, regardless of the scale of the race. The size and scope of a risk assessment should be determined by the hazards recognised, not the public profile of the event. [2.6.2]
13. The response to the accident on the water was a good example of how a safety team should be organised and react in an emergency. [2.6.3]
14. Competitors and officials should not take actions which introduce unnecessary hazards into racing. [2.6.4]
15. The rules created for UKFFORC and ORC classes to race together were confusing and did not promote safer driving. [2.6.4]
16. Mandatory training and assessment is required to develop the skills and measure the competence of all powerboat racers. [2.6.5]
17. Volunteer race officials must be supported with clear guidelines and regular training seminars to keep them updated on the authority to impose safety constraints. [2.6.5]
18. Careful review of all classes of powerboat racing boats is required to thoroughly explore and evaluate ways of improving crew protection and minimising racing hazards. [2.6.6]
19. Approval of new racing classes should be dependent on demonstration that safety has been measurably improved. [2.6.6]
20. Standards of helmet safety must be kept under continuous review and only the most appropriate helmets worn by competitors. [2.6.7]